

Fraud360

Fraud Risk Management & Early Warning Signals Platform

A multi-tenant, RBI-aligned FRMS with real-time transaction monitoring, an enforced Red Flagged Account lifecycle, and regulatory reporting — delivered as a shared service to commercial banks, co-operative banks and NBFCs.

DOCUMENT	VERSION	DATE	STATUS	COMPANION
BRD / FRMS-EWS	1.3	4 August 2026	For review	HLD v1.2

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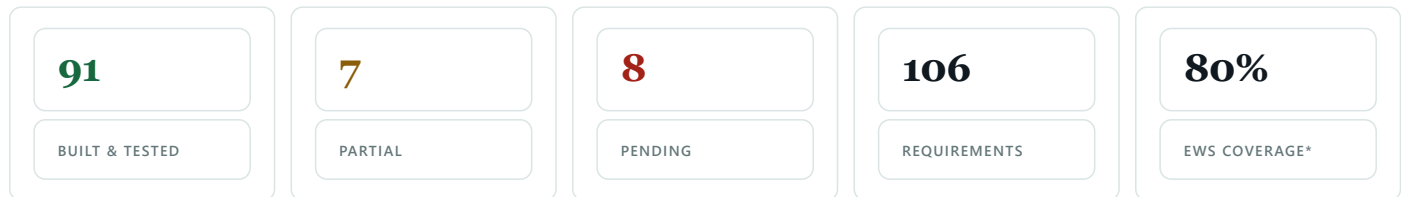
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1. Purpose and scope

What this document commits the product to, and the boundary of the first commercial release.

This BRD states the business requirements for a commercially licensed Fraud Risk Management System (FRMS) incorporating an Early Warning Signal (EWS) framework, built to the RBI Master Directions on Fraud Risk Management issued 15 July 2024 and the supervisory expectations that surround them.

The platform is **multi-tenant by design**. One deployment serves many regulated entities, each with its own branding, its own board-approved thresholds, its own data and its own users. This is a deliberate commercial position: the compliance machinery an EWS framework requires is largely identical across institutions, while the thresholds and governance around it are emphatically not.



*Coverage of *our own* 25 indicators, which is a narrower claim than coverage of RBI's illustrative list — see the note below. 20 of 25 quantitative indicators are measurable once a tenant sends its feeds: 13 from the payment stream alone, CPT-02 and CPT-03 from screening lists, and BEH-02, BEH-03 and CBS-01/02/03 from the core-banking event intake. The remaining five need records the platform cannot yet take — invoice lines, session geolocation and trade-finance documents — and are reported as unmeasurable *with the reason*, never scored as zero. See section 7.

On the RBI early-warning-signal list. The 2016 Master Directions on Frauds carried an illustrative annexure of **42** early-warning signals (widely reported as 45). The **2024** Master Directions on Fraud Risk Management (RBI/DOS/2024-25/118, 15 July 2024) **supersede** the 2016 Directions and **enumerate no list at all** — they require each bank's board-approved policy to define its own signals, attach EWS/RFA at the CRILC threshold of **₹3 crore**, and examine an alert within **30 days**.

Mapped against the 2016 annexure, the platform covers **12 of 42** signals, partially covers **3**, and does not cover **27**. That is not a defect. **26 of the 27** are borrower-conduct and credit-monitoring signals — audited financials, stock-audit findings, ROC filings, auditor's reports — which are read from a loan file on a periodic cycle, not from a payment stream in real time. Conversely **16 of our 25** indicators have no counterpart on that list, because they are payment-fraud typologies answering a different question. Closing the remaining 27 is a separate periodic-assessment capability (Lane C on the roadmap), not a threshold change. The mapping is generated and kept current by `docs/rbi_ews_mapping.py`.

IN SCOPE

- Transaction ingestion across UPI, IMPS, NEFT, RTGS and card rails
- **Inline decisioning (Lane A)** — a go/no-go answer inside the payment window for rails a tenant configures for it, alongside the existing queue-drained detection lane. Shipped in *shadow* mode: every decision is computed, recorded and reportable, and the enforced answer stays `allow` until a tenant turns enforcement on. See BR-312 to BR-315
- Rule-based EWS detection driven by each entity's configured catalogue
- Alert triage, case management and the full RFA lifecycle including natural justice
- Regulatory reporting to RBI (FMR) and FIU-IND (STR/CTR)
- Board, supervisory, investigative and inspection-facing analytics
- Tenant onboarding, branding, configuration and user administration

EXPLICITLY OUT OF SCOPE FOR RELEASE 1

- KYC/CKYC onboarding, AML customer risk rating and sanctions screening as primary systems — the platform consumes these as feeds
- Cheque truncation, trade finance document examination and loan origination
- Recovery, legal case management and LEA case tracking beyond referral records

2. Market and business case

Why a shared platform, and who buys it.

The July 2024 Directions extended a materially uniform fraud-governance obligation across three very different populations of regulated entity. A large commercial bank can fund a bespoke FRMS. A Tier-1 urban co-operative bank, a state co-operative bank or a mid-size NBFC carries the same obligations — board-approved policy, EWS framework, RFA classification, natural justice, FMR reporting — against a fraction of the technology budget. That asymmetry is the market.

SEGMENT	GOVERNING INSTRUMENT	SUPERVISOR	COMMERCIAL POSITION
Commercial banks, incl. RRBs, and AIFIs	MD on Fraud Risk Management (Commercial Banks & AIFIs), Jul 2024	RBI	Displace or augment an incumbent FRMS; strongest on EWS coverage evidence and reconciliation
Urban co-operative banks (Tiers 1–4)	MD on Fraud Risk Management (Co-operative Banks), Jul 2024	RBI	Primary target. Obligation without budget; tiered thresholds are a product feature
State & central co-operative banks	MD on Fraud Risk Management (Co-operative Banks), Jul 2024	NABARD	Same as UCB, with NABARD reporting lines
NBFCs and HFCs	MD on Fraud Risk Management (NBFCs), Jul 2024	RBI / NHB for HFCs	Growing enforcement attention; lighter transaction volume, same lifecycle burden

CENTRAL PRODUCT CLAIM

Every figure the platform reports is derived from a single metric definition and is proven to reconcile at query time. A board pack, a supervisory return and an analyst's screen cannot disagree, because they are the same SQL. This is the differentiator against dashboards assembled from independent queries, and it is verifiable rather than asserted.

3. Regulatory basis

The instruments that generate the obligations traced in section 15.

REF	INSTRUMENT	BEARING ON THE PLATFORM
REG-01	RBI Master Directions on Fraud Risk Management (three instruments), 15 Jul 2024	Board-approved fraud risk policy; EWS framework; RFA classification; staff accountability; FMR reporting; governance committee structure
REG-02	<i>State Bank of India v. Rajesh Agarwal</i> (SC, 2023)	A borrower must be given a hearing and a reasoned order before being classified as fraudulent. Procedural, not advisory
REG-03	Prevention of Money Laundering Act and FIU-IND reporting rules	STR and CTR obligations; principal officer role; filing timelines
REG-04	RBI FREE-AI Committee report, Aug 2025	Model governance, explainability, human oversight, model inventory and drift monitoring where AI/ML is used
REG-05	Digital Personal Data Protection Act, 2023	Purpose limitation, data minimisation, retention limits, processor obligations
REG-06	RBI MD on Outsourcing of IT Services, Apr 2023	Directly governs this platform as a service provider: right to audit, exit, concentration risk, sub-contracting, data localisation
REG-07	RBI storage of payment system data directive, Apr 2018	Payment data to be stored within India
REG-08	RBI MD on IT Governance, Risk, Controls and Assurance Practices, Nov 2023	Change management, access control, audit logging, BCP/DR

THRESHOLD VALUES ARE THE ENTITY'S, NEVER THE VENDOR'S

Response windows, filing deadlines, materiality limits and severity bands differ by entity type and tier, and each must be the figure that entity's own board approved. The platform ships documented defaults so a new tenant is never unconfigured, and treats every one of them as a value the tenant must review and own. No number in this document should be read as regulatory advice.

4. Stakeholders and personas

Eleven roles, each with a different question and a different permitted view of the same data.

PERSONA	PRIMARY QUESTION	MAY UNMASK CUSTOMER PII
Fraud Analyst (L1)	Which alerts in my queue are real?	Yes, with justification
Senior Investigator (L2)	What is the full picture behind this case?	Yes, with justification
Fraud Risk Manager	Are we within SLA and correctly staffed?	Yes, with justification
Principal Officer (PMLA)	What must be filed with FIU-IND, and by when?	Yes, with justification
Compliance Supervisor	Are the regulatory clocks being met?	Yes, with justification
Board / Executive	What is our fraud exposure and trend?	No — aggregate only
CRO / ACB / Special Committee	Is the framework itself working?	No — aggregate only
Model Risk / Data Science	Is detection still fit for the population?	No — models run on masked data
Internal Audit / RBI Inspection	Can each figure and decision be evidenced?	Read-only, with justification
Tenant Administrator	Is my institution configured correctly?	Yes, with justification
Platform Administrator (vendor)	Is the service healthy across tenants?	No — see below

THE OUTSOURCING BOUNDARY

Vendor staff cannot unmask a bank's customer identifiers under any role. Operating the service does not require seeing the customers, and REG-06 makes this the first question an outsourcing review asks. It is enforced in the permission model and in the database grants, not by policy alone.

5. Requirements — tenancy, onboarding and configuration

BR-1xx. Making one platform serve many regulated entities, each governed by its own board.

ID	REQUIREMENT	PRIORITY	STATUS
BR-101	Onboard a regulated entity with slug, legal name, entity type and, for UCBs, supervisory tier	MUST	BUILT
BR-102	Apply tenant branding — display name, logo, primary/accent/neutral colours, default theme, custom domain	MUST	BUILT
BR-103	Derive the tenant's default FRM policy from entity type and tier, including governing Direction and reporting lines (RBI / NABARD / NHB / FIU-IND)	MUST	BUILT
BR-104	Allow a tenant administrator to review and override every policy threshold, recording that it is board-approved	MUST	BUILT
BR-105	Version every configuration change; activate one version at a time; retain superseded versions	MUST	BUILT
BR-106	Provide a structured rule editor so thresholds can be retuned without editing JSON	SHOULD	BUILT
BR-107	Suspend, resume and offboard a tenant, with a complete data export on exit (REG-06)	MUST	BUILT
BR-108	Invite tenant users with a role, a temporary password and forced reset at first login	MUST	BUILT
BR-109	Self-service tenant onboarding and an isolated sandbox tenant for integration testing	SHOULD	PENDING
BR-110	Meter usage per tenant for billing (transactions ingested, alerts scored, seats)	SHOULD	BUILT
BR-111	Gate the seeded EWS rule catalogue by entity-type capability — a licence that cannot extend credit (Payments Bank) does not get the 9 credit-linked rules (loan-account misuse, borrowal-account conduct, collateral, qualitative credit signals) seeded at onboarding	MUST	BUILT

6. Requirements — transaction ingestion

BR-2xx. Getting real payments into the platform without corrupting them on the

way in.

ID	REQUIREMENT	PRIORITY	STATUS
BR-201	Accept transactions over an authenticated API in native rail formats for UPI, IMPS, NEFT, RTGS and card	MUST	BUILT
BR-202	Map each rail to one canonical transaction shape at the edge, so detection is rail-agnostic	MUST	BUILT
BR-213	Authenticate bank systems (payment switch, CBS extract, ETL) with a machine credential that has no interactive login path, a scope that cannot reach customer data, optional certificate binding, an address allow-list and enforced expiry — rather than a user account with MFA disabled	MUST	BUILT
BR-203	Represent money as integer paise end to end; reject amounts finer than one paisa rather than rounding	MUST	BUILT
BR-204	Reject timestamps without a timezone offset, with an explanatory error	MUST	BUILT
BR-205	Deduplicate on the rail's own identifier so a replayed file does not double-count value	MUST	BUILT
BR-206	Accept a batch partially: report each rejected row with its index and reason, and retain the rest	MUST	BUILT
BR-207	Retain the inbound payload immutably, separate from derived facts, with the time of receipt	MUST	BUILT
BR-208	Expose queue depth, provenance and freshness for operational monitoring	MUST	BUILT
BR-209	Accept bulk files over SFTP in CSV and ISO 20022, with the same validation contract	MUST	BUILT
BR-210	Support historical backfill and controlled replay after a rule change, without duplicating alerts	MUST	BUILT
BR-211	Ingest non-payment signals — CBS account events, loan account conduct, RM-raised qualitative indicators	MUST	BUILT
BR-212	Consume reference feeds: sanctions and negative lists, CERSAI charge registry, device/session telemetry	MUST	BUILT
BR-214	Ingest loan-origination (LOS) / underwriting-stage signals — falsified applications, straw borrowers, collusive valuations — not only post-disbursement CBS loan conduct. ~92% of FY25 reported bank fraud value is loan/advances fraud (RBI Annual Report FY25); today's ingestion only sees it after disbursement	SHOULD	PENDING

7. Requirements — EWS detection

BR-3xx. The indicator framework itself, and the rule that the entity's own configuration governs it.

ID	REQUIREMENT	PRIORITY	STATUS
BR-301	Maintain an EWS indicator catalogue across nine families: velocity, structuring, behavioural, layering/mule, counterparty, channel/device, trade-based, loan-account misuse, qualitative	MUST	BUILT
BR-302	Detection must read thresholds from the entity's active configuration, so retuning a band in the console changes what fires with no code change	MUST	BUILT
BR-303	A badly-tuned threshold must genuinely miss fraud — no indicator may be force-fired to populate a screen	MUST	BUILT
BR-304	Record on every alert the band matched, the observed value, the threshold applied, the unit, and the configuration version in force at scoring time	MUST	BUILT
BR-305	Score both the paying and the receiving account, so a collection account that only receives is still assessed	MUST	BUILT
BR-306	Report indicators that cannot be measured for want of a data feed, with the reason, and never approximate them	MUST	BUILT
BR-307	Maintain a dormant-indicator register: configured indicators that produced no signal in the period	MUST	BUILT
BR-308	Group related alerts into one case by account so a mule ring is one investigation, not forty alerts	MUST	BUILT
BR-309	Detect within seconds of receipt, sustained, at each tenant's peak transaction rate	MUST	BUILT
BR-310	Support network/graph typologies requiring multi-hop traversal (circular flow, layered chains)	MUST	BUILT
BR-311	Allow an authorised user to simulate a threshold change against historical data before activating it	SHOULD	BUILT
BR-312	Return a go/no-go decision (allow, challenge, hold, decline) inside the payment window, on a per-rail latency budget set by the tenant	MUST	BUILT
BR-313	Run inline decisioning in shadow mode first — compute and record the decision that <i>would</i> have been taken while returning allow, so a tenant can measure the effect before enforcing it	MUST	BUILT
BR-314	Route each rail to the inline lane or the near-real-time lane by tenant configuration, and refuse a request for a rail configured for the other lane rather than answering allow	MUST	BUILT

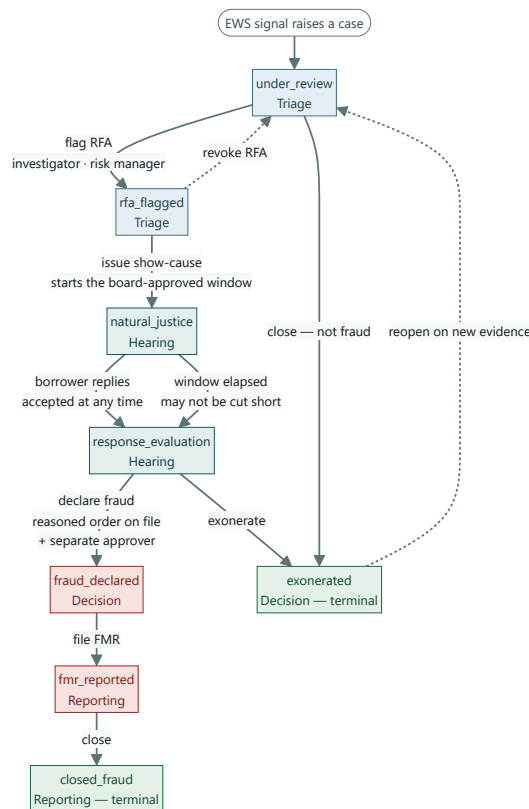
ID	REQUIREMENT	PRIORITY	STATUS
BR-315	When the latency budget expires mid-evaluation, report which rules were skipped and record the outcome as budget-exceeded — never as a clean pass	MUST	BUILT

CURRENT COVERAGE, STATED PLAINLY

Eleven indicators are computed from the payment stream today, giving 96% coverage of the configured quantitative catalogue in the reference tenant. Eight further indicators are inert pending the feeds in BR-211 and BR-212 and are reported as such. Coverage should be quoted to prospects with that qualification attached, not as a headline percentage.

8. Requirements — alert triage and the RFA lifecycle

BR-4xx. What happens after an indicator trips — the part the Directions are mostly about.



The RFA lifecycle as enforced. The spine runs top to bottom through the four stages — **triage**, **hearing**, **decision**, **reporting** — shaded by stage. Solid edges advance a case; dotted edges are the two permitted reversals. Every transition is gated by role, required evidence and elapsed time. **closed_fraud** is terminal and cannot be reopened: it has been reported to the regulator.

ID	REQUIREMENT	PRIORITY	STATUS
BR-401	Case state may change only through the defined lifecycle; no other path may write it	MUST	BUILT
BR-402	Each transition is permitted only to specified roles	MUST	BUILT
BR-403	Issuing a show-cause notice starts the borrower's response window at the entity's board-approved length	MUST	BUILT
BR-404	The response window may not be cut short by any user at any seniority (REG-02)	MUST	BUILT
BR-405	A borrower's reply may be recorded at any time and moves the case forward	MUST	BUILT
BR-406	Fraud may not be declared without a reasoned order attached to the case (REG-02)	MUST	BUILT

ID	REQUIREMENT	PRIORITY	STATUS
BR-407	Declaring fraud requires a maker and a separate checker; self-approval is refused	MUST	BUILT
BR-408	An action taken outside its policy window is permitted but permanently recorded as a breach	MUST	BUILT
BR-409	Every attempt, including refused ones, is written to an append-only case history	MUST	BUILT
BR-410	Unavailable actions are shown disabled with the reason, never hidden	SHOULD	BUILT
BR-411	Cases may be assigned to a named investigator; assignment is audited	MUST	BUILT
BR-412	Attach evidence documents to a case with a content hash; documents cannot be deleted	MUST	BUILT
BR-413	Notify the assignee, and escalate on approaching or breached clocks, by email and in-app	MUST	BUILT
BR-414	Record staff accountability examination outcomes against a declared fraud (REG-01)	MUST	BUILT
BR-415	Record LEA referral and recovery progress against a closed fraud	SHOULD	BUILT

9. Requirements — regulatory reporting

BR-5xx. The filings the lifecycle exists to produce.

ID	REQUIREMENT	PRIORITY	STATUS
BR-501	Track FMR and STR due dates from the decision date, using the entity's configured filing windows	MUST	BUILT
BR-502	Classify every fraud against an FMR category	MUST	BUILT
BR-503	Report filings due, filed and overdue, per period, reconciled to case detail	MUST	BUILT
BR-504	Generate the FMR submission artefact in the prescribed format, ready for upload	MUST	PARTIAL
BR-505	Generate the STR artefact for FIU-IND in the prescribed format	MUST	PARTIAL
BR-506	Retain every submitted artefact with its generation timestamp and the data version behind it	MUST	BUILT
BR-507	Produce a board / ACB pack on a schedule, distributed to named recipients	MUST	BUILT
BR-508	Support CTR aggregation and threshold reporting	SHOULD	PENDING
BR-509	Report fraud incidents to RBI's Central Fraud Registry (CFR) — the nine 2026 Directions each dedicate a Chapter VI sub-section to it, distinct from FMR. Likely an extension of BR-504/505's filing-format binding rather than a new cross-tenant sharing mechanism; tenant isolation is not to be weakened for this	SHOULD	PENDING

10. Requirements — analytics and assurance

BR-6xx. Thirteen role-scoped dashboards over one metric definition.

ID	REQUIREMENT	PRIORITY	STATUS
BR-601	Define every metric exactly once; all dashboards read that definition	MUST	BUILT
BR-602	Run reconciliation invariants at query time and show the result beside the figures	MUST	BUILT
BR-603	Provide role-scoped dashboards for each persona in section 4	MUST	BUILT
BR-604	Filter on 24 dimensions including rail, family, severity, state, product, segment, region, FMR and STR status	MUST	BUILT
BR-605	Drill from any figure through to the individual transaction and its evidence trail	MUST	BUILT
BR-606	Show the rule-evaluation trace: why this fired, on what observation, under which config version	MUST	BUILT
BR-607	Visualise the counterparty network with hub identification	SHOULD	BUILT
BR-608	Trend every headline metric over daily, weekly and monthly buckets	MUST	BUILT
BR-609	Save, share and schedule views; distribute exports to named recipients	SHOULD	PARTIAL
BR-610	Export to CSV with a watermark and a row ceiling, audited	MUST	BUILT

ID	REQUIREMENT	PRIORITY	STATUS
BR-611	Distinguish live traffic from demonstration data in every view	SHOULD	BUILT

11. Requirements — identity, access and privacy

BR-7xx. The largest concentration of unmet requirements, and the hardest blocker to a bank IS sign-off.

ID	REQUIREMENT	PRIORITY	STATUS
BR-701	Enforce role-based access across modules and dashboards	MUST	BUILT
BR-702	Mask customer identifiers by default; unmasking requires capability plus a recorded justification	MUST	BUILT
BR-703	Audit every view of customer-level evidence, not only every change	MUST	BUILT
BR-704	No tenant may read another tenant's data under any role or query path	MUST	BUILT
BR-705	Maintain an immutable platform audit trail of every action, actor, role and outcome	MUST	BUILT
BR-706	Force password change at first login; enforce password complexity	MUST	BUILT
BR-707	Multi-factor authentication for all users	MUST	BUILT
BR-708	Federated sign-on against the bank's own IdP (SAML / OIDC), with group-to-role mapping	MUST	BUILT
BR-709	Session timeout, refresh tokens, revocation and concurrent-session control	MUST	BUILT
BR-710	Account lockout and credential-stuffing protection on authentication	MUST	BUILT
BR-711	Per-caller rate limiting at the platform edge	MUST	BUILT
BR-712	Encryption at rest with tenant-scoped key management; TLS in transit	MUST	PARTIAL
BR-713	Retention schedule and automated purge, aligned to fraud-record retention and DPDP minimisation (REG-05)	MUST	BUILT
BR-714	All payment data stored within India (REG-07)	MUST	PARTIAL
BR-715	Maker-checker on privileged configuration changes, not only on fraud declaration	SHOULD	PENDING

12. Requirements — model and detection governance

BR-8xx. FREE-AI expectations, applied to a rules engine today and to models when they arrive.

ID	REQUIREMENT	PRIORITY	STATUS
BR-801	Maintain an inventory of every detection version that has scored live traffic	MUST	BUILT
BR-802	Flag any configuration version scoring live traffic that is not currently approved	MUST	BUILT
BR-803	Monitor population drift (PSI, KS) against a reference window and band the result	MUST	BUILT
BR-804	Log every human override of a detection outcome with its rationale	MUST	BUILT
BR-805	Every alert must be explainable to a non-technical reviewer without reference to source code	MUST	BUILT
BR-806	Backtesting harness: replay a candidate configuration against history and compare outcomes	MUST	BUILT
BR-807	Champion/challenger evaluation before a configuration is promoted	SHOULD	PENDING
BR-808	Record model approval by the responsible committee before activation	SHOULD	PENDING

13. Requirements — platform operations

BR-9xx. What the vendor must be able to evidence under REG-06 and REG-08.

ID	REQUIREMENT	PRIORITY	STATUS
BR-901	Each service owns its data; no service may read another's tables	MUST	BUILT
BR-902	Run multiple replicas of every service behind a load balancer without inconsistent results	MUST	BUILT
BR-903	A failing component must not take down healthy ones	MUST	BUILT
BR-904	Client retries must not duplicate an action	MUST	BUILT
BR-905	Distinguish liveness from readiness so a dependency blip does not restart the fleet	MUST	BUILT
BR-906	Correlate a request across every service it touches	MUST	BUILT
BR-907	Metrics, tracing and log shipping to a central observability stack	MUST	PARTIAL
BR-908	Documented and tested backup, restore and disaster recovery, with stated RPO and RTO	MUST	PENDING
BR-909	Capacity and load testing evidence at each tenant's contracted peak rate	MUST	PENDING
BR-910	Right-to-audit support: evidence pack, control descriptions, sub-processor register (REG-06)	MUST	PENDING
BR-911	Documented exit plan with data return in a usable format and agreed deletion	MUST	PARTIAL

14. Non-functional requirements

Targets to be contracted per tenant. Those marked unvalidated have not been measured and must not be quoted commercially until they are.

ID	ATTRIBUTE	TARGET	STATUS
NFR-01	Detection latency	Under 5 seconds from receipt to alert at peak	UNVALIDATED
NFR-02	Ingestion throughput	Sustained peak TPS per tenant, contracted individually	UNVALIDATED
NFR-03	Dashboard response	95th percentile under 3 seconds on 12 months of data	UNVALIDATED
NFR-04	Availability	99.5% monthly for the console; 99.9% for ingestion	DESIGN SUPPORTS
NFR-05	Recovery	RPO 15 minutes, RTO 4 hours	PENDING
NFR-06	Data residency	All data in Indian regions (REG-07)	DEPLOYMENT-DEPENDENT
NFR-07	Retention	Fraud records for the statutory period; operational data minimised (REG-05)	PENDING
NFR-08	Auditability	Every figure reproducible as at any past date, against the config then in force	BUILT
NFR-09	Accessibility	WCAG 2.1 AA for the console	NOT ASSESSED
NFR-10	Browser support	Current Chrome, Edge, Firefox, Safari; no external CDN dependency (air-gapped deployments)	BUILT

15. Obligation traceability

Each regulatory obligation to the requirements that discharge it, and the honest state of each.

OBLIGATION	SOURCE	REQUIREMENTS	STATE
Board-approved fraud risk policy governs thresholds	REG-01	BR-103, BR-104, BR-105	MET — ACTIVATION RECORDS THE APPROVING BODY, DATE AND MINUTE REFERENCE

OBLIGATION	SOURCE	REQUIREMENTS	STATE
EWS framework with defined indicators	REG-01	BR-301, BR-307	MET
EWS operates on actual transaction data	REG-01	BR-201 – BR-208, BR-302	MET — RAILS, CBS EVENTS AND REFERENCE FEEDS
Red Flagged Account classification and review	REG-01	BR-401, BR-402	MET
Hearing before adverse classification	REG-02	BR-403, BR-404, BR-405	MET
Reasoned (speaking) order for a fraud finding	REG-02	BR-406, BR-412	MET
Fraud declaration under dual control	REG-01	BR-407	MET
FMR reporting within the prescribed window	REG-01	BR-501, BR-502, BR-503, BR-504	CLOCKS TRACKED; ARTEFACT ASSEMBLED, VALIDATED AND RETAINED — PRESCRIBED FORMAT PENDING OD-04
STR reporting to FIU-IND	REG-03	BR-505	ASSEMBLED AND RETAINED; PRESCRIBED FORMAT PENDING OD-04
Staff accountability examination	REG-01	BR-414	MET
Board and ACB oversight reporting	REG-01	BR-507, BR-603	MET — DASHBOARDS AND SCHEDULED PACK WITH RECORDED DELIVERY
Model governance, explainability, human oversight	REG-04	BR-801 – BR-805	MET FOR A RULES ENGINE
Model validation before deployment	REG-04	BR-806, BR-807, BR-808	BACKTESTING BUILT; CHAMPION/CHALLENGER AND COMMITTEE APPROVAL PENDING
Access control, MFA, privileged access management	REG-08	BR-701, BR-707 – BR-710	MET — RBAC, MFA, OIDC/SAML, SESSIONS, LOCKOUT
Purpose limitation and data minimisation	REG-05	BR-702, BR-703, BR-713	MET — MASKING, RETENTION FLOORS, AUTOMATED PURGE
Payment data localisation	REG-07	BR-714	DEPLOYMENT-DEPENDENT

OBLIGATION	SOURCE	REQUIREMENTS	STATE
Service-provider audit, exit and concentration risk	REG-06	BR-107, BR-910, BR-911	EXIT EXPORT RETURNS THE FULL FRAUD RECORD AND DECLARES ITS OWN COMPLETENESS; RIGHT-TO-AUDIT PACK PENDING
Change management and audit logging	REG-08	BR-105, BR-705, BR-409	MET

16. Assumptions, constraints and exclusions

Assumptions

- Each tenant can expose transaction data by API or file within minutes of settlement
- Each tenant's board approves its own thresholds; the vendor supplies defaults, never advice
- Banks will federate identity from an existing IdP rather than manage users in the platform long-term
- Reference data (sanctions, CERSAI, CBS extracts) can be licensed or supplied by the tenant

Constraints

- Payment data must remain in India, constraining hosting and any sub-processor
- Some tenants require air-gapped or private-cloud deployment; no runtime dependency on an external CDN
- Money is integer paise throughout; no floating-point representation is permitted anywhere in the stack
- Reported figures must reconcile exactly, which rules out eventually-consistent read models for reported metrics

Exclusions

- The platform is not a system of record for customers, accounts or the general ledger
- It does not authorise, block or reverse payments
- It does not provide legal advice on classification; it enforces the process around the decision

17. Release plan

Sequenced by what blocks a signed contract, not by engineering convenience.

RELEASE	THEME	CONTENTS	EXIT CRITERION	STATUS
R1	Bankable	BR-707 – BR-710 (MFA, SSO, sessions, logout); BR-413 (notifications); BR-712, BR-713 (encryption, retention)	A bank's IS and security review can be passed	NEARLY MET MFA, SSO (OIDC + SAML), sessions, logout, notifications and retention are built, and the secrets a database reader could abuse are sealed. The one remaining item is the infrastructure half of BR-712 : full-disk or tablespace encryption, which the application cannot provide.
R2	Filable	BR-504 – BR-506 (FMR and STR artefacts); BR-414 (staff accountability); BR-507 (scheduled board pack)	A full fraud can be taken from signal to submitted return inside the platform	NOT MET Staff accountability, artefact retention and board-pack production and delivery are all built. It is not met for one reason: BR-504 and BR-505 require the <i>prescribed</i> RBI and FIU-IND formats, and the platform declares both bindings <code>not_bound</code> pending OD-04 . That is a compliance decision, not engineering work.
R3	Complete detection	BR-209 – BR-212 (files, backfill, CBS, reference feeds); BR-310 (graph typologies); BR-309 (streaming latency)	EWS coverage no longer qualified by missing feeds	NEARLY MET File intake now covers ISO 20022 as well as CSV, and backfill and replay, graph typologies, CBS events and scheduled reference feeds are all built. Only BR-309 remains - sub-second detection at contracted peak is asserted rather than measured, and closing it needs the load-test evidence of BR-909 .

RELEASE	THEME	CONTENTS	EXIT CRITERION	STATUS
R4	Operable at scale	BR-907 – BR-909 (observability, DR, load testing); BR-910, BR-911 (audit and exit); NFR validation	NFR targets measured and contractable	PARTLY MET Metrics, request correlation and pipeline-freshness gauges are in place. Tracing and log shipping, DR (BR-908), load-test evidence (BR-909) and the right-to-audit pack (BR-910) are not.
R5	Commercial	BR-109, BR-110 (self-service, metering); BR-806 – BR-808 (backtesting, champion/challenger)	Onboarding a tenant needs no vendor engineering	PARTLY MET Backtesting (BR-806) and per-tenant usage metering (BR-110) are built. A tenant administrator can now configure their own identity providers from the console — create, edit and remove OIDC/SAML sources, with role-mapping validated against the tenant's own assignable roles — closing the specific gap that previously forced vendor engineering for IdP setup. The console can also now create and activate a tenant's FRM policy version directly — <code>ConfigCreate.kind</code> was refusing <code>policy</code> at the schema; that is fixed, and BR-104 's threshold override and the Principal Officer designation (BR-504/505) are reachable end-to-end rather than only by writing to the database. Self-service tenant onboarding itself (BR-109) is still not built, and user invitation remains a platform-only action (<code>POST /tenants/{id}/users</code> requires <code>platform_admin</code>), so a new tenant still needs vendor engineering to exist in the first place.

18. What still blocks a commercial release

The honest list, ordered by what stops a signature soonest. Everything here is a MUST that a bank, its IS reviewer or its regulator asks about directly.

REF	GAP	WHY IT BLOCKS	GATE
BR-504/505	FMR and STR artefacts in the <i>prescribed</i> submission format	The single blocker on R2. The platform assembles, validates, hashes and retains the return and refuses to generate one with gaps, but the format bindings are declared <code>not_bound</code> . A named officer can submit from the pack; the platform cannot claim the return is in RBI's or FIU-IND's own format until OD-04 is decided. This needs the published schemas, not engineering. Separately: a defect that left the STR path permanently unfileable — the Principal Officer field had no source anywhere in the platform, so every STR was refused regardless of format — is fixed and verified end-to-end against a live case, and a tenant admin can now set that field themselves through the console (below), with the designation gated by the same board-attestation mechanism as every other policy fact.	R2
BR-712	Full-disk / tablespace encryption	Application-layer sealing of MFA seeds, IdP secrets and feed credentials is done. Encrypting the volume itself protects a stolen disk or a discarded backup and is a deployment control — Postgres TDE, LUKS or the cloud provider's volume encryption. An IS review asks for both.	R1
BR-909	Capacity evidence beyond a single worker	Detection latency is now measured — 200 tps per worker at p99 2.65 s, with the run refusing to call itself sustained if the backlog climbs. What is still unevidenced is multi-worker scaling and the managed-database deployment. The machine-to-machine credential that previously blocked driving the intake path under load is now built (BR-201), so this is a matter of running the test at scale rather than a missing capability.	R4
BR-908	Tested backup, restore and disaster recovery with stated RPO/RTO	Contractual. An untested restore is an assumption, and the outsourcing directions expect the numbers to be evidenced.	R4
BR-910	Right-to-audit evidence pack and sub-processor register	RBI's outsourcing directions give the regulated entity and the RBI a right to audit the service provider. It has to be answerable on request.	R4
BR-907	Distributed tracing and log shipping	Metrics, structured logs and a request id that crosses every service are in place. Nothing ships them to a central stack, so an incident is investigated by reading files on each host.	R4
BR-211 (residual)	Invoice, geolocation and trade-finance records	Five indicators — SME-03, CHN-02 and TBM-01/02/03 — still cannot be measured. They are reported as unmeasurable with the reason rather than scored, which is correct, but it is EWS coverage a bank is paying for and not receiving.	R3
BR-214	Loan-origination (LOS) / underwriting-stage signals	Ingestion today is post-disbursement only (CBS loan conduct). ~92% of FY25 reported bank fraud value is loan/advances fraud, so a falsified application or a straw borrower is invisible to the platform until after money has moved. Timing not yet committed to a release.	Unscheduled
BR-509	Central Fraud Registry (CFR) reporting	Each of the nine 2026 Directions carries a Chapter VI sub-section on CFR reporting, distinct from FMR/STR. Not built and not yet scoped in detail; most likely rides the same format-binding work as BR-504/505 rather than requiring new cross-tenant infrastructure. Timing not yet committed.	Unscheduled

19. Open decisions

Items requiring a business decision before the corresponding engineering can be specified.

REF	DECISION	WHY IT MATTERS	OWNER
OD-01	Deployment model: shared multi-tenant, dedicated schema, or single-tenant install per bank	Larger banks may refuse shared infrastructure; changes cost model and DR design	Commercial
OD-02	Whether case management becomes a separately deployed service	Correct bounded context, but requires event sync that trades exact reconciliation for eventual consistency	Product + Engineering
OD-03	Whether to sit in the authorisation path for interdiction — decided: yes , per-tenant and per-rail. Built as Lane A in shadow mode; what remains open is the narrower question of which tenant is the first to <i>enforce</i>	Settled the latency budget (50–150 ms by rail), the availability class and the liability profile. Competitive necessity: Clari5, Signzy and Servosys all answer in the payment window	Executive — closed
OD-04	Which FMR and STR submission formats to target first	Formats differ by entity type and channel; determines R2 scope	Compliance
OD-05	Reference data: licensed centrally by the platform, or supplied per tenant	Sanctions and CERSAI licensing cost and liability	Commercial
OD-06	Whether ML-based detection is in roadmap scope	Triggers the full FREE-AI model-governance obligation set, not just the rules subset	Executive

Status summary. 106 business requirements: **91 built**, **7 partial**, **8 pending**. Statuses reflect verified behaviour in the current build — every item marked Built is covered by an automated test in a suite of 809. Partial always means the requirement is not yet met as written; the reason is stated in section 17b.

This document describes a product under development. It is not legal or regulatory advice; each regulated entity remains responsible for its own compliance and for the threshold values it approves.